

Shuqi Zhao

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SUMMARY

My research interest and previous work mainly focus on robotics manipulation, especially in **dexterous hand manipulation**. In my previous research, I've gained extensive experiences on **1)** transferring human demonstration to dexterous hand manipulation (learning from human demonstration), **2)** zero-shot sim-to-real transfer for complexed contact-rich dexterous hand manipulation, and **3)** developing reasoning algorithms for error and failure analysis for dexterous manipulation leveraging LLM. The primary goal of my research endeavors lies in achieving more precise and more generalizable manipulation skills on dexterous hands with high human/robot data efficiency and high quality of sim-to-real transfer. Besides dexterous manipulation, I've also worked on robot manipulation skills generalization and had some former projects related to deep learning, control theory and optimization. With ample experiences in PyTorch, C++, ROS, Isaac Gym/Lab and Pybullet, I have gained plenty experiences about large language model, transformer, reinforcement learning, control theory and optimization in both simulation and real world.

EDUCATION

Zhejiang University, Hangzhou, Zhejiang, China

September 2019 – June 2023

College of Control Science and Engineering

- Major in automation and control science; Minor in Advanced Honor Class of Engineering Education
- Overall GPA: 3.98/4.00, 90.34/100, Major GPA: 3.99/4.00, Integrated Rank: 6/106

University of California, Berkeley, California, the United State

September 2023-Present

Control area in Department of Mechanical Engineering, second-year Ph.D.

- Supervisor: Prof. Masayoshi Tomizuka

MAIN RESEARCH EXPERIENCE

DexCtrl: Towards Sim-to-Real Dexterity with Adaptive Controller Learning.

University of California, Berkeley

Supervisors: Prof. Masayoshi Tomizuka

- Identify the mismatch of robot controllers as a critical factor in the sim-to-real gap and propose a novel method to adjust the control parameters adaptively.
- Design a simple and elegant framework to jointly obtain actions and controller parameters through separate models based on historical information, which offers better adaptivity to force variation.
- Extensive experiments on two different tasks show our method can significantly outperform baselines in both simulation and the real world, along with thorough analysis of what control parameters impact.

DexH2R: Task-oriented Dexterous Manipulation from Human to Robot

University of California, Berkeley

Supervisors: Prof. Masayoshi Tomizuka

- A tracking policy for dexterous hand that directly transfer human hand motions into dexterous hand actions, aiming at following human hand motions as well as finishing grasping tasks.
- Final actions are combined with two parts: Primitive actions computed by human hand poses through retargeting and residual actions obtained through a goal-conditioned reinforcement policy, taking in both human and object demonstrations.
- Outperformed other dexterous hand tracking policy greatly in both unseen and seen objects. Can be transferred on real robots.

CARF: Contrastive Attraction–Repulsion of Failure-Guided Flow Matching

University of California, Berkeley

Supervisors: Prof. Masayoshi Tomizuka

- A failure-guided framework that leverages imperfect robot demonstrations for more data-efficient policy learning.
 - Introduces an attraction–repulsion flow-matching objective to learn from useful behaviors while avoiding failure-critical actions, outperforming imitation learning baselines in simulation and real-world tasks.
 - Extensive experiments on two different failure types (policy rollout & teleoperation error) show our method can outperform baselines in both simulation and the real-world tasks, along with thorough analysis of how failure actions design takes impact.
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SELECTED PUBLICATIONS

- B. Du*, Y. Xie*, **S. Zhao***, Y. Chen, M. Wu, and M. Tomizuka, “One from Infinity: Actualizing Futures from Pretrained World Models into Robot Actions”, arXiv preprint arXiv: 2609.36413, Under Review
- **S. Zhao**, B. Du, C. Wu, Y. Xie, Y. Wang, and M. Tomizuka, “CARF: Contrastive Attraction-Repulsion of Failure-Guided Flow Matching”, arXiv preprint arXiv: 2609.21982, Under Review
- **S. Zhao**, K. Yang, Y. Chen, C. Li, Y. Xie, X. Zhang, C. Wang, and M. Tomizuka, “DexCtrl: Towards Sim-to-Real Dexterity with Adaptive Controller Learning,” ICRA 2026.
- **S. Zhao***, X. Zhu*, Y. Chen, C. Li, X. Zhang, M. Ding, and M. Tomizuka, “Dexh2r: Task-oriented dexterous manipulation from human to robots,” arXiv preprint arXiv:2411.04428, IEEE/ASME Transactions on Mechatronics (TMech).
- K. Xu, **S. Zhao**, Z. Zhou, Z. Li, H. Pi, Y. Zhu, Y. Wang, and R. Xiong, “A joint modeling of vision-language-action for target-oriented grasping in clutter,” in 2023 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2023, pp. 11 597–11 604.
- **S. Zhao**, D. Sun, J. Zhang, H. Lu, Y. Wang, R. Xiong, and K. Grattan, “Actuation and biomedical development of micro-/nanorobots—a review,” Materials Today Nano, vol. 18, p. 100223, 2022.

MAIN INTERNSHIP EXPERIENCES

Transfer human demonstrations to robot motions without any real robot demonstration. **Amazon IRG, North Reading**
Supervisors: Dr. Mustafa Mukadam. May 2026 – August 2026

- Investigate which aspects of zero-shot dexterous manipulation from human demonstrations can truly be addressed using a purely vision-based approach, and to what extent this can eliminate the need for robot demonstrations.
- Leverage robot-centric training data through human-hand removal, motion retargeting, and robot embodiment inpainting as transfer training policy
- Demonstrates that human-only training can transfer task semantics and coarse spatial reasoning to real robots, while precise hand–object interaction remains the main challenge for zero-shot dexterous transfer.

Developing reasoning algorithms for error and failure analysis for dexterous manipulation **Honda, San Jose**
Supervisors: Dr. Rana Soltani Zarrin. May 2025 – August 2025

- Leverage and finetune LLM to infer error and failure analysis for dexterous manipulation
- Combined tactile number and 3D point cloud occlusion and enhance LLM’s ability to understand such information

Design and control of a specially designed 7-Dof redundant robot arm **CERN, Switzerland**
Supervisors: Dr. Mario Di Castro and Dr. Eloise Matheson November 2022 – May 2023

- Implemented low-level actuators communication, control theory and camera realization for the redundant robot arm using C++.
- Implemented and modified the inverse kinematics for redundant manipulator using C++ (cooperated with co-workers). By using this, the trajectory of end effector can satisfy certain requirements such as collision avoidance.
- Conducted real-world experiments on the 7-DoF robot arm and ensured stability of all algorithms above.

HONOR & AWARD

- **Departmental First Year Doctoral Fellowships awarded in 2023 by Department of Mechanical Engineering in University of California, Berkeley.**
- Student Scholarship awarded in 2020 and 2022 by Zhejiang University
- Honorable Mention of Mathematical Contest in Modeling in 2021, rewarded by COMAP (Consortium for Mathematics and Its Applications)
- First-Class of Chinese University Intelligent Robot Creative Competition in Zhejiang University, Second-Class in Zhejiang Province

OTHERS

- Teaching assistant of Advanced Control Theory in Zhejiang University